

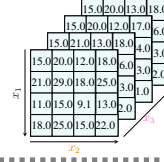
The multivariate Loewner Framework (mLF)

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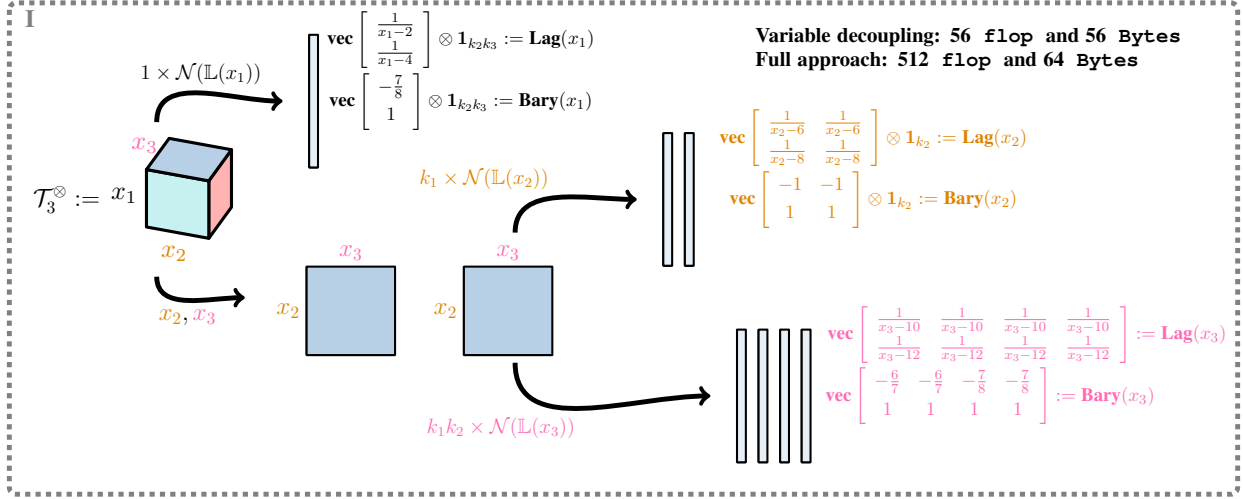
- (i) Allows n -variable rational (approximate) functions fitting from any n -D tensor or n -variate function.
- (ii) Provides a numerically robust solution to the Kolmogorov Superposition Theorem (KST) for rational functions.
- (iii) Bridges "Approximation theory" with "Systems theory", and connects with KAN.
- (iv) Allows to treat tensor problems where the data form a full hypercube by means of the matrix framework.



$$\mathbf{H}(x_1, x_2, x_3) = \frac{x_2(x_1 + 1)}{x_1 + 1} \Rightarrow \mathcal{T}_3^\otimes \in \mathbb{R}^{4 \times 4 \times 4} \text{ along } (x_1, x_2, x_3)$$



Variable decoupling

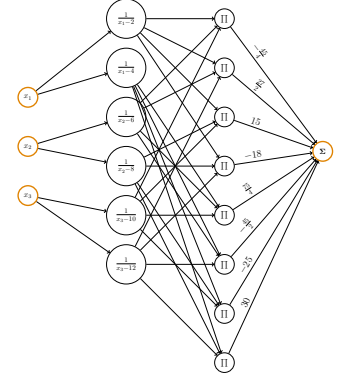


KST-like rational form

II

$$\mathbf{G}_{\text{kst}}(x_1, x_2, x_3) = \frac{\sum_{\text{rows}} \mathbf{w} \odot \Phi_1(x_1) \odot \Phi_2(x_2) \odot \Phi_3(x_3)}{\sum_{\text{rows}} \Phi_1(x_1) \odot \Phi_2(x_2) \odot \Phi_3(x_3)}$$

$$\Phi(x_1) = \begin{bmatrix} \frac{7}{-8(x_1-2)} \\ \frac{7}{-8(x_1-2)} \\ \frac{7}{-8(x_1-2)} \\ \frac{7}{-8(x_1-2)} \\ \frac{1}{x_1-4} \\ \frac{1}{x_1-4} \\ \frac{1}{x_1-4} \\ \frac{1}{x_1-4} \end{bmatrix} \quad \Phi(x_2) = \begin{bmatrix} \frac{1}{-x_2-6} \\ \frac{1}{-x_2-6} \\ \frac{1}{x_2-8} \\ \frac{1}{x_2-8} \\ \frac{1}{x_2-8} \\ \frac{1}{x_2-6} \\ \frac{1}{x_2-6} \\ \frac{1}{x_2-8} \end{bmatrix} \quad \Phi(x_3) = \begin{bmatrix} \frac{6}{-7(x_3-10)} \\ \frac{6}{-7(x_3-10)} \\ \frac{6}{-7(x_3-10)} \\ \frac{6}{-7(x_3-10)} \\ \frac{1}{8(x_3-10)} \\ \frac{1}{-x_3-12} \\ \frac{1}{-x_3-12} \\ \frac{1}{-x_3-12} \end{bmatrix} \quad \mathbf{w} = \begin{bmatrix} 15 \\ 108 \\ 20 \\ 144 \\ 150 \\ 45 \\ 200 \\ 30 \end{bmatrix}$$

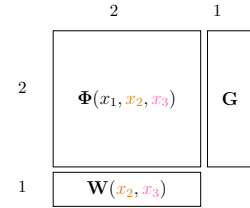


Realization

III

$$\mathbf{G}_{\text{real}}(x_1, x_2, x_3) = \mathbf{W}(x_2, x_3) \Phi(x_1, x_2, x_3)^{-1} \mathbf{G}$$

$$\Phi(x_1, x_2, x_3) = \begin{bmatrix} x_1 & -1.0 \\ 1.0 x_3 & 1.0 \end{bmatrix} \quad \mathbf{W}(x_2, x_3)^\top = \begin{bmatrix} 1.0 x_2 x_3 \\ 1.0 x_2 x_3 \end{bmatrix} \quad \mathbf{G} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$



LFR uncertain form

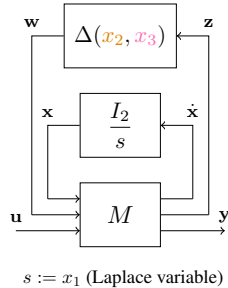
IV

$$\mathbf{G}_{\text{lfr}}(x_1, x_2, x_3) = \mathbf{M}_{21} \Delta (E - \mathbf{M}_{11} \Delta)^{-1} \mathbf{M}_{12} + \mathbf{M}_{22}$$

$$\mathbf{M} = \begin{bmatrix} \mathbf{M}_{11} & \mathbf{M}_{12} \\ \mathbf{M}_{21} & \mathbf{M}_{22} \end{bmatrix} \quad \mathbf{M}_{11} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \end{bmatrix} \quad \mathbf{M}_{12} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad \mathbf{M}_{21}^\top = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \\ 0 \end{bmatrix} \quad \mathbf{M}_{22} = 0$$

$$\mathbf{E} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \quad \Delta = \begin{bmatrix} \frac{I_2}{x_1} & 0 \\ 0 & \Delta(x_2, x_3) \end{bmatrix} = \begin{bmatrix} \frac{1}{x_1} & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{x_1} & 0 & 0 & 0 \\ 0 & 0 & x_2 & 0 & 0 \\ 0 & 0 & 0 & x_3 & 0 \\ 0 & 0 & 0 & 0 & x_3 \end{bmatrix}$$

Differential operator $x_1 := s$ and parameters are x_2, x_3 .



$s := x_1$ (Laplace variable)

>> A. C. Antoulas, I. V. Gosea and C. Poussot-Vassal, "On the Loewner framework, the Kolmogorov superposition theorem, and the curse of dimensionality", in SIAM Review (Research Spotlight), Vol. 67(4), pp. 737-770, November 2025.

>> C. Poussot-Vassal, I. V. Gosea, P. Vuillemin and A. C. Antoulas, "Tensor-based multivariate function approximation: methods benchmarking and comparison", <https://arxiv.org/abs/2506.04791>.